

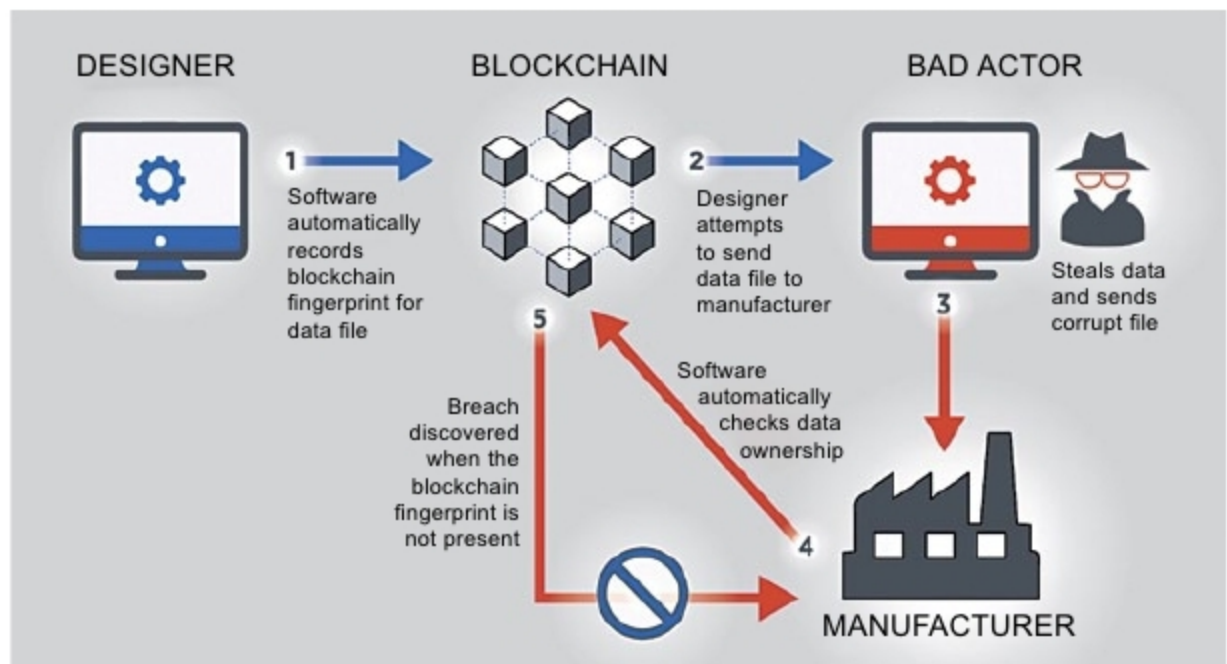
Potential of blockchain technologies

Food loss will diminish greatly, and the produce that ends up in consumers' carts will be fresher when blockchain technology, IoT devices and AI algorithms will be implemented together. Food supply chains from farmers to grocery suppliers will improve as each participant in the supply chain will know exactly how much to plant, order and ship.

Biometrics-based data enhances a multitude of health services using blockchain technology. Core services of the programme include training for medical professionals, better patient records and communication, and greater ability to track a mother's progress during pregnancy. Blockchain technology ensures that these services are delivered efficiently and cost-effectively, beginning with digital medical records to digital transfer of supporting health documents and integration with other health programmes. Overall communication between patients and medical institutions will create a value transfer platform that can be used to receive medicine.

Blockchain-based payment service providers are developing an open platform to enable blockchain developers and businesses to list digital tokens within XPOS, XWallet and XPASS. Zac Cheah, co-founder and chief executive officer, Pundi X, explains, "XPASS card can be considered as a physical crypto wallet that allows customers to top up and use their digital assets easily. Through XWallet, users can easily manage their digital assets, check current balances or top up the supported digital assets in XWallet. By default, all registered XWallet users will have their own virtual XPASS card in the app."

Digital transactions require secure yet user-friendly solutions. This needs blockchain applications where user credentials are one of the most critical security aspects of the system. NFC smart cards based on a security controller help blockchain system designers to easily integrate hardware-based security and quickly develop prototypes. Security tokens allow users to securely generate their unique public and private key pairs, authenticate with PIN, and sign their data and digital transactions.



How the blockchain identifies a bad actor in the system (Credit: googleimages.com)

User credentials stored in the security controller are protected from remote attacks. These attacks can be on software, micro-architecture or physical in case the token is stolen or lost.

Using blockchain technology to record an infected person's information on a distributed ledger can allow access to stakeholders in different countries. A centralised system to upload infection and infected party-related data is the fastest and most efficient way of reporting it, within and across the border. But security is a major concern for any healthcare data, especially when it is about an individual who is infected by a communicable disease. Blockchain technology, with built-in security, will be able to address these concerns with required confidentiality.

Challenges

A blockchain is a decentralised digital ledger system of data in the form of a chain of blocks that are cryptographically linked to the previous one. Though convenient, storing blockchain user credentials on a computer or cellphone can be extremely risky—an attacker may identify and read sensitive information such as the highly confidential private key of the user. Similarly, if a user forgets or loses the credentials, usage of the assets is lost, as credentials, typically, cannot be centrally-restored.

A public blockchain cannot be anonymised and/or structurally deleted from the shared ledger. Therefore the business will be unable to resolve its needs to keep records with its obligations to comply with privacy laws.

So, any business operating using a public blockchain must maintain a copy of the entire blockchain. To increase access security, token-based hardware security must be integrated into the application that can protect from software and physical attacks alike.

Future development areas

The regulatory and policy making bodies have started to support and promote blockchain growth both in India and globally. Telangana government has launched Blockchain District in India to create a global talent hub that can unlock the value of blockchain technology in democratising data and solving big-scale problems. This is also a huge step in reskilling and right-skilling the workforce of the future towards making India the blockchain capital of the world.

The Indian government has realised the potential applicability of blockchain technology to several use cases such as land registry, education and health records, and supply chain management. Education is the first step towards educating aspiring talents to develop on blockchain and drive the adoption of the ecosystem. Academies like EMURGO will be established to train developers and create blockchain experts to expand the usage of blockchain technology.

Blockchain is gradually becoming a vital part of our infrastructure, like the Internet. It will help empower all organisations and developers to build global solutions embracing blockchain technology towards a better future and a more connected world. **EFY**